

GOOGLE CLOUD RAPID AGENT HACKATHON · ARIZE TRACK

# AuditLens

The **compliance layer for AI agents**. It audits production LLM traces against the EU AI Act with Gemini — and writes the verdict back to Arize Phoenix, with a human in the loop.



Gemini



Vertex AI Agent Engine



Arize Phoenix MCP



Cloud Run



AuditLens

# Compliance can't keep up with AI

- The **EU AI Act** (Reg. (EU) 2024/1689) is in force — every LLM in the EU must prove **transparency (Art.50)**, **risk management (Art.9)**, **human oversight (Art.14)**, and **GPAI obligations (Art.53)**.
- Today that's a **quarterly, ~3-day manual review** by experts — slow, costly, and stale the day it ships.
- Penalties reach **€35M or 7% of global turnover**. Nobody is watching the traces in between.

# A daily, automated, 8-minute audit

## 1 · Collect

Real production traces from Arize Phoenix.

## 2 · Score

Each trace vs the EU AI Act rubric with Gemini.

## 3 · Roll up

Violations + the exact prompt fix for the worst article.

## 4 · Propose

Writes the verdict back — **gated on human approval**.

## 5 · Loop

Approved violations → dataset → prompt A/B experiment.

It's an **agent, not a chatbot** — a multi-step pipeline with a real approval gate.



# One service. Two agents. A human gate.

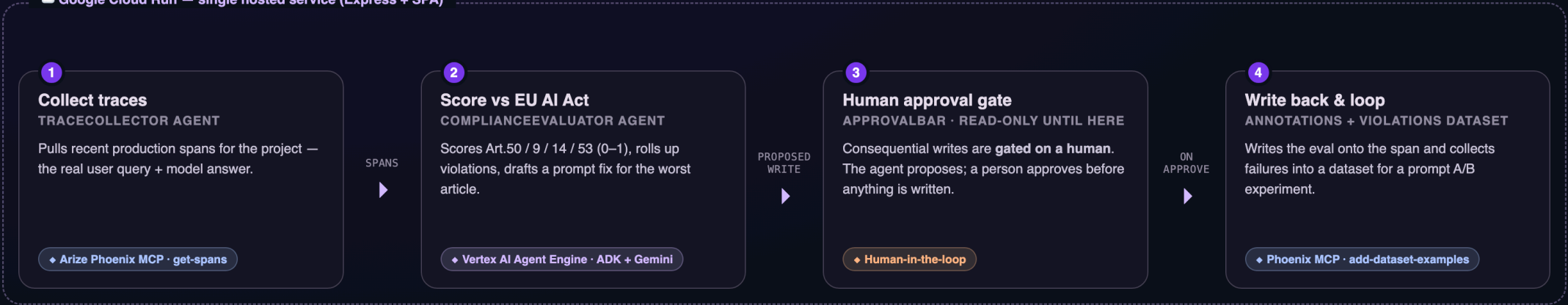


## AuditLens — Architecture

EU AI Act trace-compliance agent · Gemini · Google Cloud Agent Builder · Arize Phoenix MCP

☁ Google Cloud Run — single hosted service (Express + SPA)

Phoenix traces every LLM call → AuditLens scores each against the EU AI Act with Gemini → and, **with your approval**, writes the verdict back & collects violations for a prompt-improvement experiment.



**K ≈ 0.65**  
COHEN'S K · SUBSTANTIAL

**Evaluator Calibration — “eval the evals”**  
The Gemini judge is itself validated live against a human-labelled golden set (20 traces · 100 judgements). We measure the judge, then improve it — that’s what makes the compliance score defensible.

- REQUIRED STACK · ALL INVOKED AT RUNTIME
- Gemini (3 + 2.5)
  - Agent Builder · Agent Engine
  - Arize Phoenix MCP
  - Cloud Run
  - Secret Manager



REQUIRED TECH · ALL INVOKED AT RUNTIME

# Not just named — actually called

| REQUIREMENT                                                                                                         | HOW AUDITLENS USES IT AT RUNTIME                                                                  |
|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|  <b>Gemini</b>                     | Gemini 3 (gemini-3-flash-preview) direct + Gemini 2.5 inside the agent — scores every trace       |
|  <b>Google Cloud Agent Builder</b> | A deployed <b>Vertex AI Agent Engine</b> agent (ADK), called per scoring request                  |
|  <b>Arize Phoenix</b>              | Spawns the <b>Phoenix MCP server</b> — reads (get-spans) <b>and</b> writes (add-dataset-examples) |

```
curl /health → agent_builder_runtime:true · partner_transport:"mcp" · model:gemini-3-flash-preview
```

# We don't trust the AI judge — we measure it

0.65

Cohen's  $\kappa$  · **substantial**

20 traces · 100 judgements · 85% acc

Anyone can have an LLM judge compliance. The reliability hole is: **how do you know the judge is right?**

AuditLens validates its own Gemini evaluator **live** against a human-labelled golden set. We measured it, found it over-flagged GPAI, **tuned it**, and  $\kappa$  rose **0.24 → 0.65**.

**“Eval the evals.”** Treat the judge like any model — measure it, then improve it. That's what makes the score defensible.



# The conformity layer for the whole platform

## Who

Any team shipping LLMs into the EU —  
fintech, health, HR, gov, support.

## Value

Continuous conformity evidence instead of  
a stale quarterly PDF; catch drift the day it  
happens; an audit trail of approvals.

## Wedge

It audits the agents you already run — point  
any OpenInference app at your Phoenix  
project and AuditLens covers the platform.

Quarterly 3-day manual review → **daily 8-minute automated audit.**

TRY IT

# The compliance layer for AI agents.



Gemini



Vertex AI Agent Engine · ADK



Arize Phoenix MCP



Cloud Run



Secret Manager



[auditlens-908307939543.europe-west1.run.app](https://auditlens-908307939543.europe-west1.run.app)



[github.com/manojmallick/auditlens](https://github.com/manojmallick/auditlens)

Open **?tour=auto** for a 90-second self-driving walkthrough.



AuditLens